UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 004 · Implicit functions, problems on implicit and composite functions · compiled 14 October 2026

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1 Short notes

1 · Implicit functions, level surfaces and the differential

A relation $f(x,y)=c$ or $f(x_1,\dots,x_n)=c$ defines an implicit function; geometrically it is a level curve or level surface (isotherm, equipotential surface), e.g. the cylinders $x^2+y^2=c$ or spheres $x^2+y^2+z^2=c$. On it f does not change, so its total differential $df=f_x\,dx+f_y\,dy+\cdots=0$. The differential is also used in substitution in integrals and in error estimates.

2 · Implicit differentiation in two variables

An implicit function is given by a relation $f(x,y)=c$, a level curve. Along it $df=f_x\,dx+f_y\,dy=0$, so $\dfrac{dy}{dx}=-\dfrac{f_x}{f_y}$ ($f_y\neq 0$). Direct differentiation of both sides gives the same result. The slope at (x_0,y_0) gives the tangent $y-y_0=m(x-x_0)$; e.g. $xe^y+ye^x=0$ has tangent $x+y=0$ at the origin.

3 · Implicit functions of three variables

For $f(x,y,z)=0$, z is a function of x and y: $\dfrac{\partial z}{\partial x}=-\dfrac{f_x}{f_z}$ (hold y), $\dfrac{\partial z}{\partial y}=-\dfrac{f_y}{f_z}$, and $\dfrac{\partial y}{\partial x}=-\dfrac{f_x}{f_y}$ (hold z), provided the denominators are non-zero. Check that the point lies on the surface. For second derivatives, remember that z still depends on x,y.

4 · Composite functions and the chain rule

If $u=f(x,y)$ and x,y depend on s,t (or r,θ), u is a composite function of s,t, and $u_s=f_x\,x_s+f_y\,y_s$, $u_t=f_x\,x_t+f_y\,y_t$: one term per intermediate variable. Write the chain first. If f is unknown, keep f_x,f_y as symbols. Eliminate x,y from the final answer. Standard results: $u_x+u_y+u_z=0$ for $f(x-y,y-z,z-x)$; $xu_x+yu_y+zu_z=0$ for $f(x/y,y/z,z/x)$.

5 · Composite functions in polar coordinates

With $x = r\cos\theta$, $y = r\sin\theta$: $x_\theta = -r\sin\theta$, $y_\theta = r\cos\theta$; with $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$: $r_x = x/r$, $r_y = y/r$, $\theta_x = -y/r^2$, $\theta_y = x/r^2$. Functions of x^2+y^2 alone have $u_\theta = 0$, $\tan^{-1}(y/x)$ has $u_\theta = 1$, and $u_x^2 + u_y^2 = u_r^2 + u_\theta^2/r^2$.

2 Keyword index

30 terms; the number in brackets is the topic.

Branch diagram (2). Diamond diagram: $f_x \cdot 1 + f_y\, dy/dx = 0$.

Chain (of a problem) (4). What depends on what; fixes the formula.

Chain rule (4). $u_s = f_x\, x_s + f_y\, y_s$.

Composite function (4). A function whose variables are functions of other variables.

Constant power rule (2). $\frac{d}{dx} a^x = a^x \ln a$.

Cylinder (1). $x^2 + y^2 = c$: a level surface for each c.

Differential (1). $df = f_x\, dx + f_y\, dy$; also called the total derivative.

Direct method (2). Differentiate both sides of $f(x, y) = c$ term by term.

Elimination (4). Replacing x, y by the new variables in the final answer.

Equipotential surface (1). Level surface of constant potential.

Error estimate (1). Application of the differential (end of the unit).

Explicit function (1). $y = g(x)$ given by a formula.

Formula method (2). $dy/dx = -f_x/f_y$ from partial derivatives.

Identity for any f (4). A result such as $u_x + u_y + u_z = 0$ that holds whatever f is.

Implicit differentiation (2). Finding dy/dx or z_x without solving the relation.

Implicit function (1). Function defined by a relation $f(x, y) = c$.

Integration by substitution (1). Uses the differential: $t = x^2$, $dt = 2x\, dx$.

Intermediate variable (4). Variable between u and the final variables.

Isotherm (1). Level surface of constant temperature.

Level curve (1). $f(x, y) = c$ in the plane.

Level surface (1). $f(x, y, z) = c$; f is constant on it.

Point on the surface (3). Must satisfy $f = 0$ before you substitute it.

Polar coordinates (5). $x = r\cos\theta$, $y = r\sin\theta$.

Polar identity (5). $u_x^2 + u_y^2 = u_r^2 + u_\theta^2/r^2$.

Second implicit derivative (3). Differentiate $-f_x/f_z$ again, remembering $z = z(x, y)$.

Slope (2). $dy/dx = -f_x/f_y$ at a point.

Sphere (1). $x^2 + y^2 + z^2 = c$: nested level surfaces.

Tangent line (2). $y - y_0 = m(x - x_0)$.

Three-variable implicit function (3). $f(x, y, z) = 0$, $z = z(x, y)$.

Variable power rule (2). $\frac{d}{dx} x^n = n x^{n-1}$.

3 Rapid recall

1	What is a level surface?	A set where $f(x,y,z)=c$; f is constant on it.
2	Name two physical level surfaces.	Equipotential surfaces, isotherms.
3	df on a level curve?	0.
4	dy/dx for $f(x,y)=c$?	$-f_x/f_y$, if $f_y \neq 0$.
5	Which way up is it?	Differentiated-by variable (x) on top: $-f_x/f_y$ (slide 004·4 is upside down).
6	dy/dx for $x^3y+3xy^2=3$?	$-3y(x^2+y)/(x(x^2+6y))$.
7	dy/dx for $x^2+xy+y^2=7$?	$-(2x+y)/(x+2y)$.
8	Tangent to $xe^y+ye^x=0$ at the origin?	$x+y=0$.
9	$\partial(\cos x)^y/\partial y$?	$(\cos x)^y \ln\cos x$.
10	dy/dx for $e^y-e^x+xy=1$ at $(1,1)$?	$(e-1)/(e+1)$.
11	$\partial z/\partial x$ from $f(x,y,z)=0$?	$-f_x/f_z$ (hold y).
12	$\partial y/\partial x$ from $f(x,y,z)=0$?	$-f_x/f_y$ (hold z).
13	$z^3+xy-y^2z=6$ at $(0,1,2)$: y_x, z_x?	$1/4$, $-1/11$.
14	First check before substituting a point?	It satisfies the equation.
15	$x^xy^yz^z=c$, z_{xy} at $x=y=z$?	$-[x\log(ex)]^{-1}$.
16	Chain rule for u_s, $u=f(x,y)$?	$f_x\,x_s+f_y\,y_s$.
17	$u=f(r)$, $r=\sqrt{x^2+y^2+z^2}$: u_x?	$f'(r)\,x/r$.
18	$u=f(x-y,y-z,z-x)$: $u_x+u_y+u_z$?	0.
19	$u=f(x/y,y/z,z/x)$: $xu_x+yu_y+zu_z$?	0.
20	$z=\log(u^2+v)$, $u=e^{x+y^2}$, $v=x+y^2$: $2yz_x-z_y$?	0.
21	x_θ, y_θ in polar form?	$-r\sin\theta$, $r\cos\theta$.
22	θ_x, θ_y?	$-y/r^2$, x/r^2.
23	$u=\sin(x^2+y^2)$: u_θ?	0.
24	$u=\tan^{-1}(y/x)$: u_θ?	1.

Mnemonic (study aid). *"Implicit: minus, the one you differentiate by on top."* $\dfrac{dy}{dx}=-\dfrac{f_x}{f_y}$, $\dfrac{\partial z}{\partial x}=-\dfrac{f_x}{f_z}$.

Mnemonic (study aid). *"Know the chain, then multiply down and add across."* One term per intermediate variable.

4 Last 10 minutes before the exam

Top facts

1. A relation $f = c$ is a level curve or surface; on it $df = f_x\,dx + f_y\,dy + \cdots = 0$.
2. Two variables: $dy/dx = -f_x/f_y$ ($f_y \neq 0$). Slide 004·4 prints it upside down.
3. Both routes (differentiate directly, or the formula) give the same dy/dx.
4. Tangent at (x_0, y_0): $y - y_0 = m(x - x_0)$ with $m = dy/dx$ there.
5. Three variables: $z_x = -f_x/f_z$ (hold y), $z_y = -f_y/f_z$, $y_x = -f_x/f_y$ (hold z).
6. Check that the point lies on the surface: the lecture's $(1, 0, 2)$ does not.
7. Composite: write the chain, then $u_s = f_x x_s + f_y y_s$; keep unknown f_x, f_y; eliminate x, y.
8. ESA favourites: $u_x + u_y + u_z = 0$; $xu_x + yu_y + zu_z = 0$; $z_x^2 + z_y^2$ identities; $2yz_x - z_y = 0$.
9. Polar: $x_\theta = -r\sin\theta$, $y_\theta = r\cos\theta$; $r_x = x/r$, $\theta_x = -y/r^2$.
10. Functions of $x^2 + y^2$ alone give $u_\theta = 0$; $\tan^{-1}(y/x)$ gives $u_\theta = 1$.

Top formulas

E1.1 · Total differential ESA ★★

$$dz = f_x(x, y)\,dx + f_y(x, y)\,dy$$

GIVES Approximate change in z for small changes dx, dy.

APPLIES $z = f(x, y)$ differentiable.

NOTE n variables: $dz = f_{x_1}dx_1 + f_{x_2}dx_2 + \cdots + f_{x_n}dx_n$.

S004·3 · S004·4 · S003·8 · T003a 00:13:56

E2.1 · Implicit differentiation, two variables ESA ★★

$$f(x, y) = c \implies \frac{dy}{dx} = -\frac{f_x}{f_y}, \qquad f_y \neq 0$$

GIVES Slope of the level curve at a point.

APPLIES $f_y \neq 0$ at the point.

NOTE From $df = f_x\,dx + f_y\,dy = 0$. Slide 004·4 prints the fraction upside down.

S004·5 · S004·6 · T004a 00:06:36 · T003b 00:10:33

E3.1 · Implicit differentiation, three variables

ESA ★★

$$f(x,y,z)=0: \quad \frac{\partial z}{\partial x}=-\frac{f_x}{f_z}, \quad \frac{\partial z}{\partial y}=-\frac{f_y}{f_z}\ (f_z\neq 0); \qquad \frac{\partial y}{\partial x}=-\frac{f_x}{f_y}\ (f_y\neq 0)$$

GIVES Partial derivatives of the implicitly defined variable.

APPLIES Hold y constant for z_x; hold z constant for y_x.

NOTE Extends to any number of variables.

S004·6 · S003·26 · T004a 00:10:23 · T003b 00:09:29

E4.1 · Chain rule, two intermediate variables

ESA ★★★

$$\frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

GIVES Partial derivatives of a composite function.

APPLIES $z=f(x,y)$, $x=g(u,v)$, $y=h(u,v)$.

NOTE All derivatives are partial, because everything depends on two variables.

S003·13 · S003·14 · T003a 00:21:07

E5.1 · Polar coordinates: derivatives you reuse

ESA ★★

$$x=r\cos\theta,\ y=r\sin\theta: \qquad x_\theta=-r\sin\theta, \quad y_\theta=r\cos\theta$$
$$r=\sqrt{x^2+y^2},\ \theta=\tan^{-1}\tfrac{y}{x}: \quad r_x=\frac{x}{r},\ r_y=\frac{y}{r},\ \theta_x=-\frac{y}{r^2},\ \theta_y=\frac{x}{r^2}$$

GIVES The inner derivatives for polar problems.

APPLIES $r=\sqrt{x^2+y^2}$, $\theta=\tan^{-1}(y/x)$.

NOTE So $u_x=f_r\frac{x}{r}-f_\theta\frac{y}{r^2}$ and $u_y=f_r\frac{y}{r}+f_\theta\frac{x}{r^2}$ for $u=f(r,\theta)$.

S004·13 · S004·15 · S004·17

E5.2 · Polar identity

ESA ★★

$$\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=\left(\frac{\partial u}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$$

GIVES The squared gradient in polar coordinates.

APPLIES $u=u(x,y)$, $x=r\cos\theta$, $y=r\sin\theta$.

NOTE Proved in W5.6.

CW U1 P15